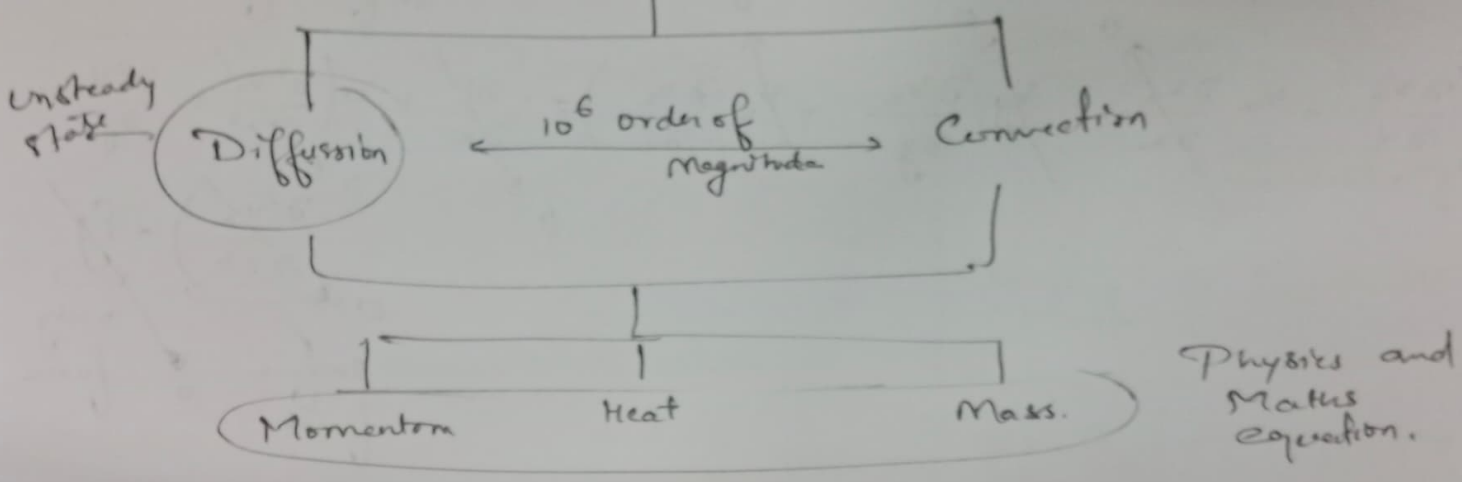
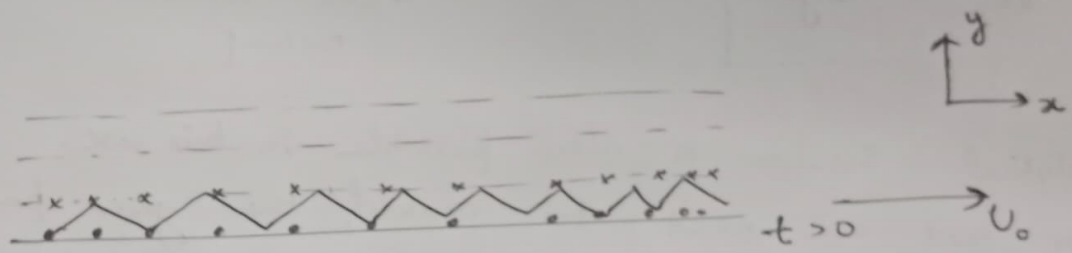
Prof. II

The phenomena of Transport



Momentum Transfer

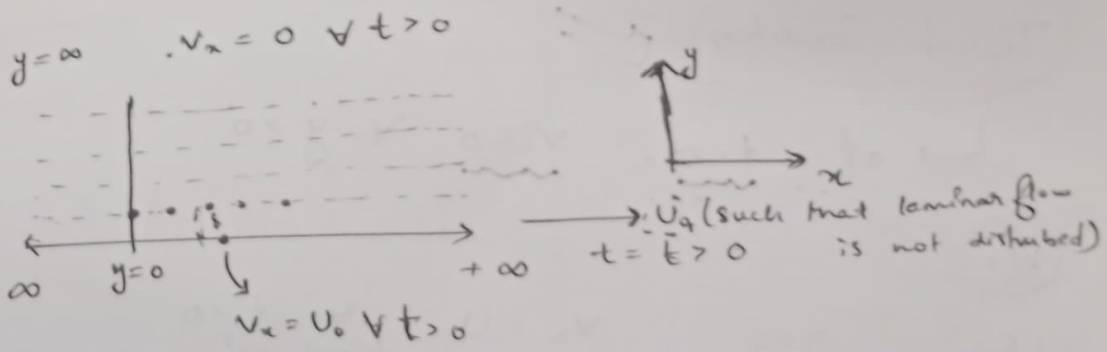


Stoke's 1st Problem

$\tau = \mu$

Assumptions

1. Unidirectional ~~stokes flow~~ motion
2. No variation is x-direction (means flow is fully developed)
3. Velocity driven flow
4. (Semi-Infinite domain)
5. fully developed flow in x and z direction $\boxed{\Rightarrow v_x \neq f(x,z)}$



To Find :- velocity profile

$F = ma$

<u>N-S equation</u>

$$\rho\left(\frac{\partial v_x}{\partial t} + \cancel{v_x \frac{\partial v_x}{\partial x}}^{\,0} + \cancel{v_y}^{\,0} \frac{\partial v_x}{\partial y} + \cancel{v_z}^{\,0} \frac{\partial v_x}{\partial z}\right) = \overbrace{\cancel{-\frac{\partial p}{\partial x}}^{\,0} + \mu\left(\cancel{\frac{\partial^2 v_x}{\partial x^2}}^{\,0} + \frac{\partial^2 v_x}{\partial y^2} + \cancel{\frac{\partial^2 v_x}{\partial z^2}}^{\,0}\right)}^{\text{surface forces}} + \cancel{\rho g}^{\,0}$$

surface forces. body force

$$\boxed{\rho \frac{\partial v_x}{\partial t} = \mu \frac{\partial^2 v_x}{\partial y^2}}$$

→ Partial differential equation

↳ can be solved by Numerical method

→ Analytical solution is also possible in this case.

→ for that we need to convert the given partial differential equation in ODE.

PDE ⇒ ODE ⇒ we need BCs (we need two BC and 1 initial condition)

<u>Initial Condition</u>

↳ at $t = 0$, $v_x = 0$ $\forall\ y > 0$

<u>BCs</u>

<u>1 B.C</u> $y = 0$, $v_x = U_0$ $\forall\ t > 0$

<u>2.B.C</u> $y = \infty$, $v_x = 0$ $\forall\ t > 0$

$$\frac{\partial v_x}{\partial t} = \frac{\mu}{\rho}\left(\frac{\partial^2 v_x}{\partial y^2}\right)$$

$\frac{\mu}{\rho}$ → Kinematic viscosity

$$\frac{\partial v_x}{\partial t} = \nu\left(\frac{\partial^2 v_x}{\partial y^2}\right)$$

ν → m^2/s

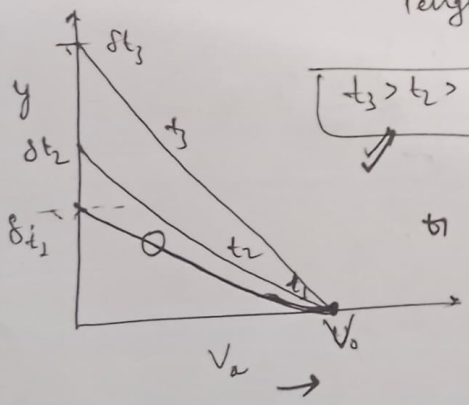
Now we need eliminate to solve so, we will club t and y together.

$\delta(t, y)$ → "Momentum diffusion length scale"

⇒ Chapter-4 of bird book

Class-II

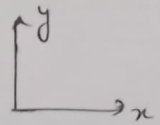
δ = Momentum diffusion length scale → fxn(y, t)

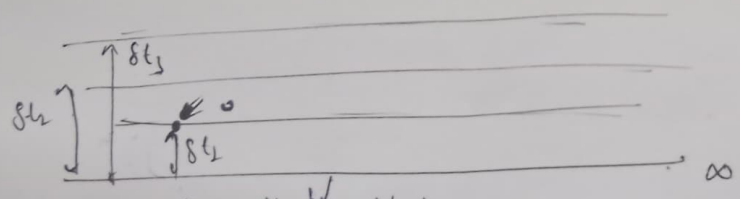


$t_3 > t_2 > t_1$

t_1

$y = \infty$, $V_x = 0$ $\forall$ $t \geq 0$





$y = 0$ $V_x = V_0$ $\forall$ $t > 0$

$t_2 > t_1$

Analogy Corresponding to ν

Heat transfer ⇒ $\alpha = \frac{k}{\rho C_p}$

k → thermal ~~Conductivity~~

Mass transfer ⇒ D_{AB}

$q = -k\frac{dT}{dx}$

$\frac{\partial T}{\partial t}$ = ⇒ Some equations will be there for mass and heat transfer as well.

Momentum Transfer

$$\frac{\partial v_x}{\partial t} = \frac{\mu}{\rho}\left(\frac{\partial^2 v_x}{\partial y^2}\right)$$

$$\frac{\partial T}{\partial t} = \alpha\left(\frac{\partial^2 T}{\partial y^2}\right)$$

$$\frac{\partial c}{\partial t} = D_{AB}\left(\frac{\partial^2 c}{\partial y^2}\right)$$

ν

all have same unit m^2/s.

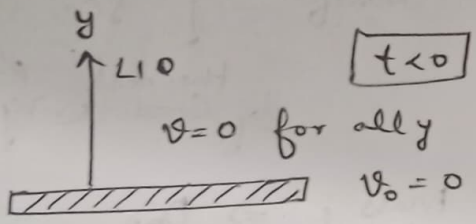
ν → momentum diffusivity

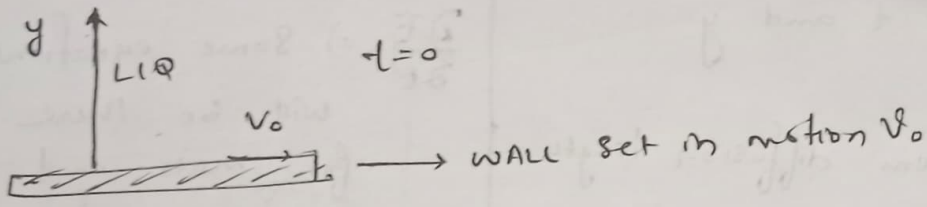
α → thermal diffusivity

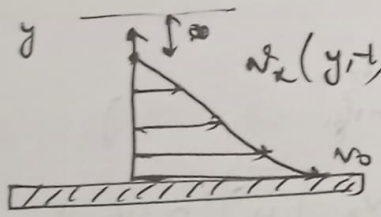
D_{AB} → mass diffusivity

"Lec-15" (Must do after)

FLOW Near a Wall Suddenly set in Motion







Unsteady flow

$$\frac{d\left(\frac{d(x)}{dt}\right)}{dt}$$

δ = order of Magnitude...

Order of Magnitude Analysis

$$\frac{\partial V_x}{\partial t} = \nu \frac{\partial^2 V_x}{\partial y^2}$$

$$\frac{\cancel{V_0}}{t} = \nu \frac{\cancel{V_0}}{\delta^2}$$

$$\delta^2 = \nu t$$

$$\boxed{\delta = \sqrt{\nu t}}$$

⇒ This method is method of combination of two variables

$$\phi = \frac{V_x}{V_0} \qquad \eta = \frac{y}{\delta_t}$$

I.C. $t = 0$, $V_x = 0$, $\phi = 0$, $\eta = \infty$

B.C. $y = 0$, $V_x = V_0$, $\phi = 1$, $\eta = 0$

$y = \infty$, $V_x = 0$, $\phi = 0$, $\eta = \infty$

$$\eta = \frac{y}{\sqrt{\nu t}} \quad \text{(Ideally)}$$

but we will use

$$\eta = \frac{y}{\sqrt{4\nu t}}$$

$$\boxed{\frac{\partial \phi}{\partial t} = \nu \frac{\partial^2 \phi}{\partial y^2}}$$

$$\frac{\partial \eta}{\partial t} = \frac{\partial \left(\frac{y}{\sqrt{4\nu t}}\right)}{\partial t}$$

$$\frac{y}{\sqrt{4\nu}} \frac{\partial t^{-1/2}}{\partial t}$$

$$-\left(\frac{1}{2}\right) t^{-3/2}$$

$$-\frac{y}{\sqrt{4\nu}} \left(\frac{1}{2}\right) \frac{1}{t\sqrt{t}}$$

$$-\frac{1}{2}\frac{\eta}{t}$$

$$\frac{y}{\sqrt{4\nu}} \frac{dt^{-1/2}}{dt}$$

$$\left(-\frac{1}{2}\right) \frac{y}{t\sqrt{4\nu}}$$

$$\frac{\partial \phi}{\partial t} = \frac{d\phi}{d\eta} \cdot \frac{\partial \eta}{dt}$$

$$\Rightarrow \frac{d\phi}{d\eta} \cdot \frac{d\left(\frac{y}{\sqrt{4\nu t}}\right)}{dt}$$

$$\frac{d\phi}{d\eta} \cdot \frac{2y\sqrt{t}}{\sqrt{4\nu}} \quad (A)$$

$$\frac{\partial \phi}{\partial t} = -\frac{d\phi}{d\eta} y \left(\sqrt{\frac{t}{\nu}}\right)$$

$$-\frac{d\phi}{d\eta} \eta \sqrt{4\nu t} \sqrt{\frac{t}{\nu}}$$

$$\boxed{\frac{\partial \phi}{\partial t} = -\frac{1}{2}\frac{\eta}{t}\frac{d\phi}{d\eta}} \quad (A)$$

$$\frac{\partial \phi}{\partial y} = \frac{d\phi}{d\eta}\frac{d\eta}{dy} = \frac{d\phi}{d\eta}\frac{\partial\left(\frac{y}{\sqrt{4\nu t}}\right)}{\partial y}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial \phi}{\partial y}\right) = \frac{\partial\left(\frac{d\phi}{d\eta}\left(\frac{1}{\sqrt{4\nu t}}\right)\right)}{\partial y}$$

$$\frac{1}{\sqrt{4\nu t}}\frac{d}{dy}\left(\frac{d\phi}{d\eta}\right) + \frac{d\phi}{d\eta}\frac{\partial}{\partial y}\left(\frac{1}{\sqrt{4\nu t}}\right)$$

$$= \frac{d\phi}{d\eta}\frac{1}{\sqrt{4\nu t}} = \frac{d\phi}{d\eta}\left(\frac{1}{\sqrt{4\nu t}} + \frac{y}{\sqrt{4\nu t}}\right)$$

$$\Rightarrow \frac{1}{4\nu t}\frac{d^2\phi}{d\eta^2}$$

$$= \frac{1}{4\nu t}\frac{d^2\phi}{d\eta^2} \quad (B)$$

$$\Rightarrow -\frac{1}{x}\frac{\eta}{t}\frac{d\phi}{d\eta} = \nu \times \frac{1}{4\nu t}\frac{d^2\phi}{d\eta^2}$$

$$\phi = \frac{V_x}{V_0}$$

$$\boxed{\frac{d^2\phi}{d\eta^2} + 2\eta\frac{d\phi}{d\eta} = 0}$$

solve this

$$\eta = \frac{y}{\sqrt{4\nu t}}$$

$$\delta = \sqrt{4\nu t}$$

of the form

$$\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 0$$

$$\frac{d^2y}{dx^2} = -2x\frac{dy}{dx}$$

$$\frac{dy}{dx} = -$$

$$\boxed{\begin{array}{ll} \text{B.C.1} & \eta = 0, \ \phi = 1 \\ & \eta = \infty, \ \phi = 0 \end{array}}$$

$$\cancel{I.F. = e^{\int 2\eta\, d\eta} = \eta^2}$$

$$I.F. = e^{\int 2x\, dx}$$

$$= e^{\int \frac{x^2}{x}\, dx}$$

$$\cancel{\phi = \int e^{-\eta^2}\, d\eta\, C_1 + C_2}$$

Let $\frac{d\phi}{d\eta} = \psi$

$$\frac{d\psi}{d\eta} + 2\eta\psi = 0 \Rightarrow \frac{d\psi}{d\eta} = -2\eta\psi$$

$$\Rightarrow \frac{d\psi}{\psi} = -2\eta\, d\eta$$

$$\ln \psi = -\eta^2 + c_1'$$

$$\psi = c_1 \exp(-\eta^2 + c_2')$$

$$\int d\phi = \int c_1 \exp(-\eta^2)\, d\eta$$

$$\boxed{\phi = c_1 \int_0^{\eta} \exp(-\eta^2)\, d\eta + c_2}$$

$\eta = 0$, $\phi = 1$

$$\boxed{c_2 = 1}$$

$\eta = \infty$, $\phi = 0$

$$c_1 \int_0^{\infty} \exp(-\eta^2)\, d\eta + 1 = 0$$

$$c_1 = -\frac{1}{\int_0^{\infty} \exp(-\eta^2)\, d\eta}$$

$$\boxed{\phi = 1 - \frac{\int_0^{\eta} \exp(-\eta^2)\, d\eta}{\int_0^{\infty} \exp(-\eta^2)\, d\eta}}$$

error function

$$\boxed{erf = \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{\pi}} e^{-t^2}\, dt.}$$

Solve this and get this

$$\boxed{\phi = 1 - erf(\eta)}$$

$$\delta = \sqrt{4\nu t} \Rightarrow \boxed{\phi = erfc(\eta)}$$

Assignment

→ Try to solve the

$$\delta = \sqrt{2\nu t}$$

$$\delta = \sqrt{\nu t}$$

→ Assignment have time of two weeks....

Solve $\int_0^{\infty} \exp(-\eta^2)\, d\eta$

$$\Rightarrow \int_0^{\infty} z^{-1/2} e^{-z}\, dz$$

$$\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} z^{1/2 - 1} e^{-z}\, dz$$

$$\Rightarrow \boxed{\frac{V_x}{U_0} = 1 - erf\left(\frac{y}{\sqrt{4\nu t}}\right)}$$

Note: Go back to some pages from other notes has been started now.

$$\boxed{\frac{v_x}{v_0} = 1 - \operatorname{erf}\left(\frac{y}{\sqrt{4\nu t}}\right)}$$

Graph

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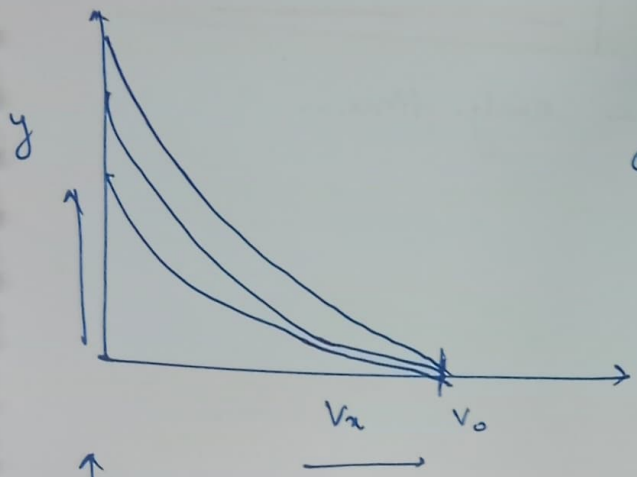

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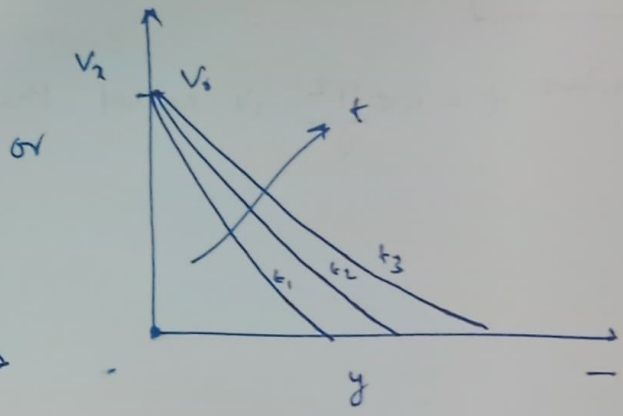

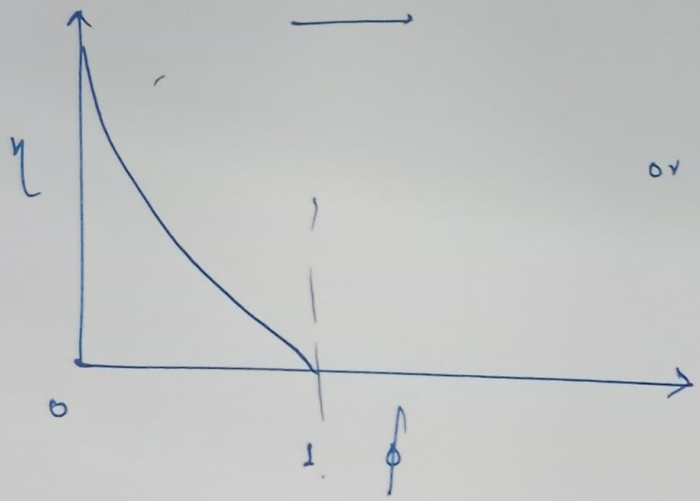

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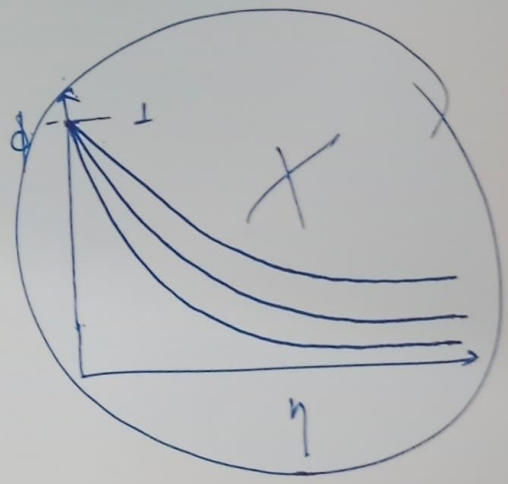
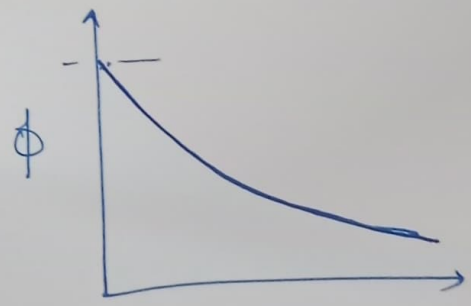


due to similarity solution...



$$\eta = \frac{y}{\sqrt{4\nu t}}$$

Semi-Infinite domain